Surya Thirukonda

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Education

Texas A&M University, Craig and Galen Brown Engineering Honors — College Station, TX
Bachelor of Science in Computer Science, Minor in Statistics; GPA: 4.0 — *May 2028*

Experience

Software Engineering Intern | *FastAPI, QuestDB, Docker* — June 2026 – Aug. 2026
DeKalb Capital Management — *Remote*

- Built portfolio analytics and stock screening on a dashboard serving 100+ equities, integrating yfinance and the Interactive Brokers API for position sync and time-weighted returns.
- Made brokerage sync idempotent and retry-safe, preventing duplicate position writes and stale reads; added validation, structured logging, rate limiting, and health checks.
- Added time-series storage and application-level caching to cut repeated queries; deployed via Railway and Cloudflare.

Founding Software Engineer | *FastAPI, React, Supabase, Docker* — Feb. 2026 – Present
Revio — *Remote*

- Launched a manuscript-formatting beta that has processed 200+ end-to-end conversions across 40+ journal profiles, owning architecture from user-flow design through production deployment.
- Prevented unauthorized edits to author wording with a constrained editing agent: deterministic orchestrator, JSON-only LLM decision layer, fix-policy registry, and content-fidelity checks.
- Built async, schema-validated DOCX/PDF processing on Gotenberg pipelines, containerized across Railway and Vercel.

Contract Software Engineer | *Node.js, Express, PostgreSQL, WebSockets, Stripe* — Feb. 2026 – June 2026
Mai Shan Yun (WaiterDock) — *College Station, TX*

- Won the Mai Shan Yun sponsor challenge at TAMUhack (40+ teams) and converted the 24-hour prototype into a four-month paid development contract.
- Synchronized order state across admin, waiter, kitchen, and customer roles with RBAC, payments, REST APIs, and real-time updates.
- Built a workforce module covering employee auth, clock-in/out tracking, live task assignment, SMTP alerts, and payroll.

Research & Projects

Undergraduate Researcher | *Python, Mem0, OpenAI API, GitHub Actions* — Aug. 2026 – Present
SUCCESS Lab (Prof. Guofei Gu), Texas A&M Department of Computer Science — *College Station, TX*

- Developed MemHeal, a defense that traces an LLM agent's harmful action to the memories behind it and repairs them claim by claim, cutting attack success from 61% to 7% while keeping 2x more benign facts than quarantine.
- Built counterfactual replay that re-runs tool-using agents without suspect memories to find the cause (0.88 vs. 0.19 precision), evaluated on 300 poisoning scenarios across 3 memory backends (incl. Mem0) and 4 LLMs.
- Shipped GitHub Actions CI/CD running 6,800+ offline tests and 8 pre-push checks with the LLM client disabled.

Tempo | *Node.js, WebSockets, React, PlayCanvas, SQLite, OpenAI API* — Sep. 2026

- Won MLH Best Use of Presage at HackRice 16 (team of 3) with a browser sports game that turns phones into motion controllers, built on a deterministic 120 Hz simulation with 300+ Vitest tests.
- Built the Presage camera-vitals integration in the Node.js agent service and SQLite persistence for movement history and bets, with fallbacks when the camera or SDK fails.

Physio | *MediaPipe, FastAPI, WebSockets, Raspberry Pi, ElevenLabs* — May 2026

- Placed 2nd overall at HackHCC 2026 (40+ teams) with a voice-first physical-therapy assistant turning at-home rehab into structured motion data.
- Built the pose-estimation and motion-analysis pipeline scoring reps, range of motion, and compensation over a 20 Hz WebSocket stream.

Southwest Airlines Baggage Revenue Challenge | *Python, SARIMAX, Streamlit* — Oct. 2025 – Dec. 2025

- Won 1st place in the Aggie Data Science Club x Southwest Airlines challenge with a SARIMAX pipeline forecasting 24 months at under 12% MAPE, presenting revenue-risk scenarios to Southwest stakeholders.

Technical Skills

Languages: Python, JavaScript, Java, C++, SQL, R
Backend/Web: FastAPI, Node.js, Express, React, Three.js, REST APIs, WebSockets, Prisma, SQLAlchemy, JWT, OAuth
Infrastructure: PostgreSQL, Supabase, Docker, CI/CD, GitHub Actions, AWS (EC2, S3), Vercel, Cloudflare, Git, Linux
Libraries/Testing: pandas, NumPy, scikit-learn, statsmodels, XGBoost, pytest, Vitest
Certification: AWS Certified Cloud Practitioner